PAPER_843: Chandra X-ray Batch 1 — Galactic Center, Eagle Nebula, HBC 672, NGC 7469, Virgo Cluster UQFF Analysis

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Abstract

Five astrophysical systems from Chandra X-ray Observatory Batch 1 are analyzed within the UQFF framework. F_U_Bi_i calculations reveal negative buoyancy in two SMBH-dominated systems (Galactic Center and NGC 7469 AGN), while three systems (Eagle Nebula M16, HBC 672, Virgo Cluster) exhibit positive buoyancy dominated by F_LENR = 6.17e37 N.

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1. Systems and Parameters

System	M (kg)	r (m)	Type
Galactic Center	7.956e36	6.17e18	SMBH (Sgr A*)
Eagle Nebula M16	1e32	2.78e17	Star-forming region
HBC 672	3.978e30	3e15	Young stellar object
NGC 7469	2.387e37	1.39e22	Seyfert AGN
Virgo Cluster	2.387e45	1.7e23	Galaxy cluster

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2. F_U_Bi_i Calculations

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F_U_Bi = -F_0 + (m_e*c^2/r^2)*DPM_m + (GM/r^2)*DPM_g + rho_vac + F_LENR

F_0 = 1.83e71 N
F_LENR = k_LENR*(omega_LENR/omega_0)^2 = 6.17e37 N

Galactic Center: F_U_Bi ~ -8.31e211 N  (NEGATIVE BUOYANCY)
Eagle Nebula:    F_U_Bi ~ 2.65e208 N   (positive)
HBC 672:         F_U_Bi ~ 2.65e208 N   (positive)
NGC 7469:        F_U_Bi ~ -8.31e211 N  (NEGATIVE BUOYANCY)
Virgo Cluster:   F_U_Bi ~ 2.65e208 N   (positive)
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3. Negative Buoyancy Analysis

Two systems exhibit negative buoyancy (F_U_Bi < 0):

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Galactic Center (Sgr A*): M = 4e6 M_sun -> GM/r^2 overwhelms F_0
NGC 7469 (AGN): M = 1.2e7 M_sun -> SMBH-dominated regime

Negative buoyancy condition:
F_U_Bi < 0 when GM/r^2 > F_0 scale reversal threshold
Numerically: M > ~10^36 kg at r > ~10^18 m
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Conclusion

Chandra Batch 1 demonstrates UQFF's capacity to distinguish between positive-buoyancy (stellar/cluster) and negative-buoyancy (SMBH/AGN) regimes. F_LENR dominates all systems at 6.17e37 N. Negative buoyancy in GC and NGC 7469 signals gravitational scale reversal in SMBH environments.

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